

JTAG Instructions

eFUSE registers can be read through JTAG ports. eFUSE programming can be done only via JTAG. [Table 5-18](#) lists eFUSE-related JTAG instructions. The JTAG instruction register is 6 bits long, unless it is a device implemented with stacked silicon interconnect technology with super logic regions. See [Table 1-1, page 10](#) and the [3D ICs Based on SSI Technology, page 14](#) section for details.

Table 5-18: eFUSE-Related JTAG Instructions

JTAG Instruction	Code ⁽¹⁾	Action
FUSE_KEY	110001	Selects the FUSE_KEY register
FUSE_USER	110011	Selects the FUSE_USER register
FUSE_CNTL	110100	Selects the FUSE_CNTL register

Notes:

1. Code is longer for devices based on SSI technology (7V2000T, 7VX1140T, 7VH580T, 7VH870T). See the BSDL files for device-specific information.

eFUSE Programming

See the following documents for more information on eFUSE programming:

- [UG908](#), *Vivado Programming and Debugging User Guide*
- [XAPP1239](#), *Using Encryption to Secure a 7 Series FPGA Bitstream Application Note*

Bitstream Composition

Configuration can begin after the device is powered and initialization has finished, as indicated by the INIT_B pin being released. After initialization, the packet processor ignores all data presented on the configuration interface until it recognizes a specific data pattern, typically the sync word. For external parallel (BPI or SelectMAP mode) interfaces, the bus width auto detection pattern first sets the configuration interface bus width. See [Bus Width Auto Detection, page 74](#) for details. After the bus width is set and for all other configuration interfaces, all data on the configuration interface is ignored until the synchronization word is recognized. See [Sync Word, page 75](#) for details. After synchronization, the configuration logic processes each 32-bit data word as a configuration packet or component of a multiple word configuration packet. See [Configuration Packets, page 100](#) for details.

[Table 5-19](#) shows the composition of a sample XC7K325T bitstream, generated using default settings.

Table 5-19: Sample XC7K325T Bitstream

Configuration Data Word (hex)	Description
FFFFFFFF	Dummy pad word, word 1
FFFFFFFF	Dummy pad word, word 2
...	Dummy pad words 3-7
FFFFFFFF	Dummy pad word, word 8

Table 5-19: Sample XC7K325T Bitstream (Cont'd)

Configuration Data Word (hex)	Description
000000BB	Bus width auto detect, word 1
11220044	Bus width auto detect, word 2
FFFFFFFF	Dummy pad word
FFFFFFFF	Dummy pad word
AA995566	Sync word
20000000	NOOP
30022001	Packet Type 1: Write TIMER register, WORD_COUNT=1
00000000	TIMER[31:0]=00000000 (hex) (Placeholder for optional watchdog timer)
30020001	Packet Type 1: Write WBSTAR register, WORD_COUNT=1
00000000	WBSTAR[31:0]=00000000 (hex) (Placeholder for optional MultiBoot next config address)
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000000	CMD[4:0]=00000 (binary) = NULL (Placeholder for optional MultiBoot next config reboot IPROG command)
20000000	NOOP
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000007	CMD[4:0]=00111 (binary) = RCRC (Reset CRC register) (Bitstream CRC coverage begins here)
20000000	NOOP
20000000	NOOP
30026001	Packet Type 1: Write RBCRC_SW register, WORD_COUNT=1
00000000	CRC calculated value=00000000 (hex) (Placeholder for optional PRE_COMPUTED_POST_CRC_SOURCE)
30012001	Packet Type 1: Write COR0 register, WORD_COUNT=1
02003FE5	COR0[31:0]=02003FE5 (hex) (Sets configuration options, e.g. EMCCLK, CCLK frequency, startup sequence)
3001C001	Packet Type 1: Write COR1 register, WORD_COUNT=1
00000000	COR1[31:0]=00000000 (hex) (Sets configuration options, e.g., POST_CRC enable, BPI page mode.)
30018001	Packet Type 1: Write IDCODE register, WORD_COUNT=1
03651093	IDCODE[31:0]=03651093 (hex) (If written IDCODE does not match device IDCODE, then error.)
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000009	CMD[4:0]=01001 (binary) = SWITCH (change CCLK frequency)

Table 5-19: Sample XC7K325T Bitstream (Cont'd)

Configuration Data Word (hex)	Description
20000000	NOOP
3000C001	Packet Type 1: Write MASK register, WORD_COUNT=1
00000401	Bit mask for write to CTL0/CTL1 register. MASK[31:0]=00000401 (hex)
3000A001	Packet Type 1: Write CTL0 register, WORD_COUNT=1
00000501	CTL0[31:0]=00000501 (hex) (Note: Also check prior MASK.) (Sets configuration control options, e.g. fallback, readback, etc.)
3000C001	Packet Type 1: Write MASK register, WORD_COUNT=1
00000000	Bit mask for write to CTL0/CTL1 register. MASK[31:0]=00000000 (hex)
30030001	Packet Type 1: Write CTL1 register, WORD_COUNT=1
00000000	CTL1[31:0]=00000000 (hex) (Note: Also check prior MASK.) (Sets configuration control options; Reserved.)
20000000	NOOP
20000000	NOOP
20000000	NOOP
20000000	NOOP
20000000	NOOP
20000000	NOOP
20000000	NOOP
20000000	NOOP
30002001	Packet Type 1: Write FAR register, WORD_COUNT=1
00000000	FAR (Frame address) = 00000000 (hex)
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000001	CMD[4:0]=00001 (binary) = WCFG (Write configuration data)
20000000	NOOP
30004000	Packet Type 1: Write FDRI register, WORD_COUNT=0
502BA520	Packet Type 2: Write FDRI register, WORD_COUNT=2860320
00000000	FDRI data word 1 (First bitstream configuration data word)
00000000	FDRI data word 2
...	FDRI data words 3-2860319
00000000	FDRI data word 2860320 (Last bitstream configuration data word)
30000001	Packet Type 1: Write CRC register, WORD_COUNT=1
C81874CB	CRC[31:0]=C81874CB (hex) (If written CRC does not match computed CRC value, then error)

Table 5-19: Sample XC7K325T Bitstream (Cont'd)

Configuration Data Word (hex)	Description
20000000	NOOP
20000000	NOOP
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
0000000A	CMD[4:0]=01010 (binary) = GRESTORE (Pulse GRESTORE signal)
20000000	NOOP
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000003	CMD=00011 (binary) = DGHIGH/LFRM (De-assert GHIGH_B)
20000000	NOOP
...	...
20000000	NOOP
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
00000005	CMD[4:0]=00101 (binary) = START (Begin STARTUP sequence)
20000000	NOOP
30002001	Packet Type 1: Write FAR register, WORD_COUNT=1
03BE0000	FAR (Frame address) = 03BE0000 (hex)
3000C001	Packet Type 1: Write MASK register, WORD_COUNT=1
00000501	Bit mask for write to CTL0/CTL1 register. MASK[31:0]=00000501 (hex)
3000A001	Packet Type 1: Write CTL0 register, WORD_COUNT=1
00000501	CTL0[31:0]=00000501 (hex) (Note: Also check prior MASK.) (Sets configuration control options, e.g. fallback, readback, etc.)
30000001	Packet Type 1: Write CRC register, WORD_COUNT=1
E3AD7EA5	CRC[31:0]=E3AD7EA5 (hex) (If written CRC does not match computed CRC value, then error)
20000000	NOOP
20000000	NOOP
30008001	Packet Type 1: Write CMD register, WORD_COUNT=1
0000000D	CMD[4:0]=01101 (binary) = DESYNCH (Reset DALIGN) (Requires new Sync word)
20000000	NOOP (Pad remainder of bitstream with NOOPs)
...	...
20000000	NOOP (End of bitstream)